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Modeling and Analytics

Agency Credit Loss Severity Drivers

Trading Quantitative Strategies

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GSE Credit Analyzer on Locus has been updated to include Freddie Mac loss severity vectors. See appendix for definitions of the new vectors.

Freddie Mac loan loss data: risk drivers for property disposition default and loss severities, comparisons with prime jumbo loans

Locus GSE Credit Analyzer was updated to include Freddie Mac loss severity vectors

Freddie Mac extended loan-level credit performance data beyond 180 days' delinquency and included actual loss components through property disposition.

Property disposition default has similar drivers as D180 default with longer seasoning curves and curing rates ranging between 20% and 40%. Key drivers of the seasoning ramp include the judicial state status.

Loan loss components include Default UPB, Delinquent Interest, Expenses, Net Sales Proceeds, MI Recoveries and Non-MI Recoveries. The drivers for loss components are as follows:

- Delinquent interest: Judicial/Non-Judicial foreclosure, servicer, WAC, months in delinquency.
- Expenses: Loan Size, Judicial/Non-Judicial, WAC, months in delinquency.
- Net Sales Proceeds: Current LTV, Loan Size, loan purpose, occupancy, age.
- MI Recoveries: MI% (LTV>80).
- Non MI Recoveries: SATO.

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Freddie Mac loan loss data and STACR transaction linked with actual loss

Freddie Mac announced its intention to issue mortgage credit risk transfer deals based on actual loss. To this end, on November 24, 2014, Freddie Mac extended loan-level credit performance data beyond 180 days' delinquency and included actual loss components through property disposition, which included approximately 16.9 million fully amortizing 30-year fixed-rate loans originated from January 1, 1999 through June 30, 2013, with monthly loan performance data through December 31, 2013.

In this analysis we define property disposition default, or simply "default" unless specified otherwise, as loans terminated through property disposition via either a foreclosure alternative group (short sale, third-party sale, charge off and note sale) or REO disposition.

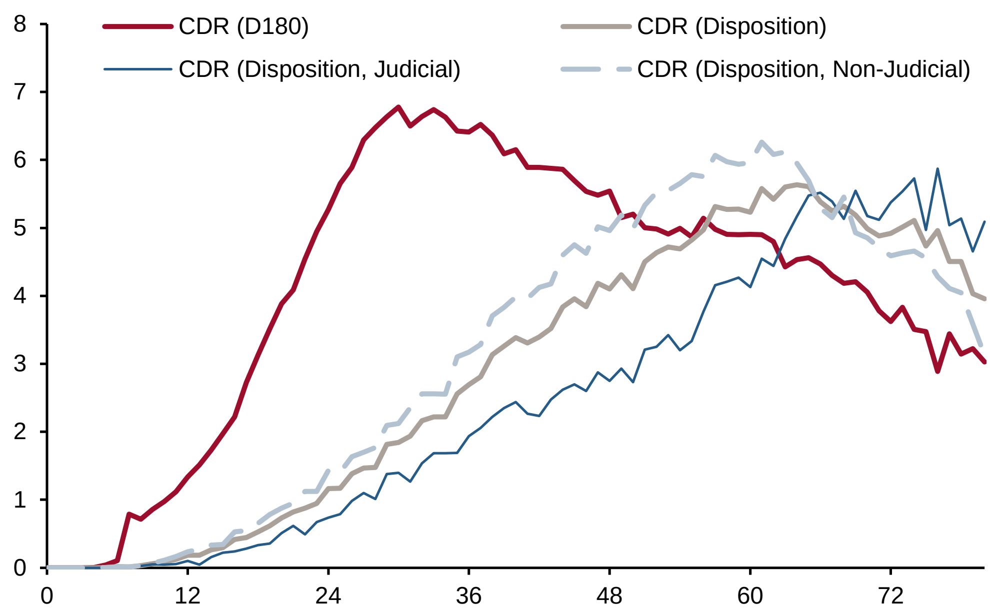
Cumulative credit performance

2005-2007 vintages have the highest default and severity

Exhibit 1 shows the comparisons between D180 and property disposition default. Property disposition default has similar drivers as D180 default, except their default seasoning curves are very different. (The D180 default drivers were discussed in detail in a previous publication. May 23, 2013 Modeling and Analytics: Fannie Mae Credit data release.) Compared to D180, the disposition default seasoning curve is pushed further along the timeline as loans moving through the liquidation pipeline. The disposition default seasoning curve is also lower than the D180 default curve, with curing rates ranging between the recent 20% and the pre-crisis level of 40%. The key driver for the extended seasoning curve include the judicial state status. The judicial foreclosure default curve ramps up much slower than the non-judicial foreclosure curve (Exhibit 1).

Exhibit 1: Default seasoning curves for D180 and property disposition defaults

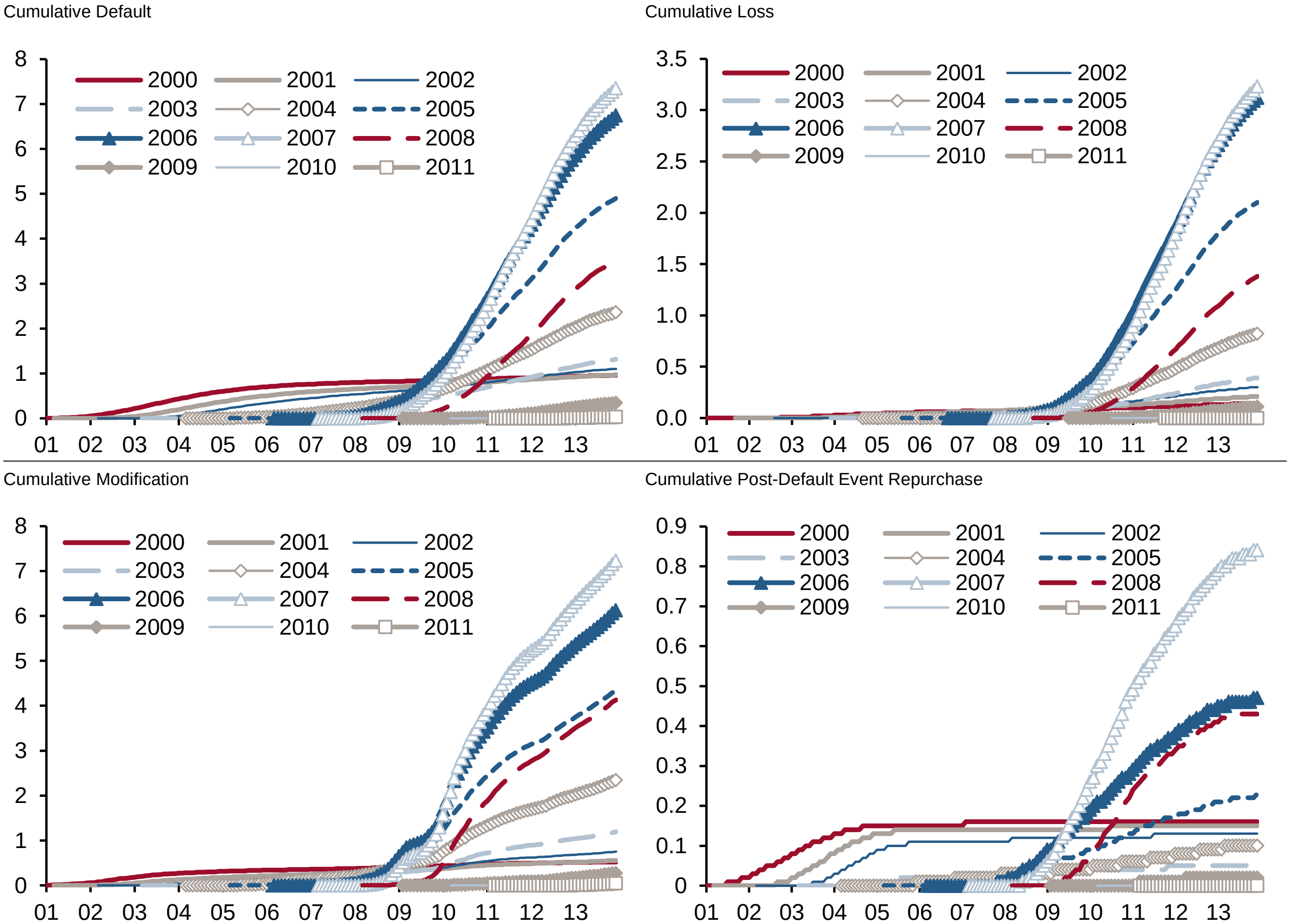
Freddie Mac 2007 Vintage



Source: Freddie Mac, Credit Suisse

Exhibit 2 shows cumulative default, loss, modification and post-default repurchase as a percentage of the original unpaid principal balance across vintages. The 2005-2007 vintages have the worst credit performance because loans originated at the peak of the housing market subsequently experienced the largest home price declines (Exhibit 3).

Exhibit 2: Cumulative performance by vintage



Source: Freddie Mac, Credit Suisse

Exhibit 3: LTVs and cumulative HPA as of Dec 31 2013 by vintage

Vintage	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
Orig LTV	78.2	75.5	73.8	72.1	73.6	72.2	72.9	74.2	71.9	67.1	69.9	71.1	71.3	72.8
Curr LTV	48.3	48.1	50.9	52.8	62.7	74.5	82.8	82.1	71.9	59.4	61.0	60.8	61.9	68.6
cumHPA	40%	32%	23%	15%	3%	**-11%**	**-15%**	**-12%**	-1%	9%	12%	16%	14%	6%

Source: Freddie Mac, Credit Suisse

Different measures of default, loss and severity

Difference in credit performance measures can be large when liquidation recovery is high absent a severe housing downturn

There are different definitions of default, loss and severity as illustrated in Exhibits 4 and 5.

The definition we use in this analysis, i.e., property disposition default, accounts for loans terminated through property disposition via either a foreclosure alternative group (short sale, third party sale, charge off and note sale) or REO disposition. If a defaulted loan went through the disposition process and recovered more than the balance owned plus all expenses, the windfall would offset losses from other loan defaults in the same group, resulting in a lower severity. This definition is used by Freddie Mac, as it measures risks Freddie Mac bears for mortgage defaults. The *Total CumDef* and *Total CumSev* in Exhibits 4 and 5 are based on this definition.

One can also examine loss severity that accounts for property dispositions only if a loss is occurred. This will result in higher severity. The *CumDef (Loss>0)* and *CumSev (Loss>0)* in Exhibit 4 and 5 are based on this definition.

The Credit Suisse non-agency model and the Locus non-agency calculators define default as a transition from delinquency to REO or as a short sale with a loss; and loss severity as loss divided by outstanding loan balance at liquidation. By this definition, foreclosure alternative disposition without loss is excluded from the default and loss severity calculations. The CS *CumDef* and CS *CumSev* in Exhibits 4 and 5 are based on this definition, except that loans still in REO are excluded. By definition, this severity measure is between *total CumSev* and *CumSev (Loss>0)*.

As illustrated in Exhibit 4 and 5, the difference in these measures were quite large prior to 2005 because of the significant percentage of delinquent loans liquidated without a loss. The differences declined for the later vintages as the housing market 'crash' made liquidation without a loss less likely.

Exhibit 4: Cumulative default and severity as of December 31 2013 (60<LTV<=80)

Vintage	Total CumDef	Total CumSev	CumDef (Loss>0)	CumSev (Loss>0)	CS CumDef	CS CumSev
1999	0.6	18.4	0.3	34.1	0.4	24.2
2000	0.5	25.5	0.4	39.1	0.4	31.1
2001	0.6	33.0	0.4	44.5	0.5	38.5
2002	0.7	39.4	0.6	48.9	0.7	44.3
2003	1.1	35.2	0.9	43.3	1.0	40.2
2004	2.2	39.8	1.9	44.6	2.0	42.7
2005	5.1	47.2	4.9	49.3	5.0	48.5
2006	7.0	50.8	6.8	52.7	6.9	52.1
2007	6.9	50.2	6.6	52.1	6.7	51.7
2008	3.3	46.3	3.1	48.6	3.2	48.1
2009	0.4	35.7	0.4	37.9	0.4	37.3
2010	0.1	29.6	0.1	32.5	0.1	31.5
2011	0.0	29.9	0.0	33.2	0.0	32.9
2012	0.0	14.7	0.0	23.6	0.0	15.5

Source: Freddie Mac, Credit Suisse

Exhibit 5: Cumulative default and severity as of December 31 2013 (80<LTV<=97)

Vintage	Total CumDef	Total CumSev	CumDef (Loss>0)	CumSev (Loss>0)	CS CumDef	CS CumSev
1999	1.7	7.0	0.8	17.4	1.4	8.9
2000	2.0	10.0	1.2	19.0	1.7	11.8
2001	2.5	14.7	1.7	23.1	2.2	16.7
2002	3.0	18.3	2.2	26.2	2.8	20.2
2003	3.4	24.3	2.7	31.3	3.1	26.4
2004	5.0	27.6	4.2	33.3	4.6	29.6
2005	9.1	32.6	8.1	36.7	8.6	34.5
2006	11.4	35.2	10.5	38.4	10.8	37.1
2007	12.6	36.0	11.7	39.0	11.9	38.2
2008	7.4	30.6	6.7	33.6	6.9	32.9
2009	1.0	18.5	0.9	21.1	0.9	20.4
2010	0.3	12.4	0.2	15.7	0.2	14.6
2011	0.1	10.5	0.1	13.1	0.1	11.9
2012	0.0	17.3	0.0	17.3	0.0	17.3

Source: Freddie Mac, Credit Suisse

Comparison between Freddie Mac and non-agency prime jumbo credit performance

Comparable default rates but different severity profile

The total cumulative default for pre-2008 vintage Freddie Mac and prime jumbo loans are comparable, however total cumulative severity for prime jumbo loans is much lower than that of Freddie Mac loans (Exhibit 6). This is because fewer prime jumbo loans liquidated with a loss compared to Freddie Mac loans, especially in the early vintages. The cumulative severity for loans defaulted with loss>0 are comparable for Freddie Mac and prime jumbo (see CumSev (Loss>0) column in Exhibits 6 and 7). Exhibits 8 and 9 show comparison of 2003 and 2007 cumulative default and severity.

Exhibit 6: Cumulative default and Severity as of December 31 2013, Freddie Mac vs prime jumbo

Vintage	Total CumDef		Total CumSev		CumDef (Loss>0)		CumSev (Loss>0)		FICO		LTV	
	FH	JF	FH	JF	FH	JF	FH	JF	FH	JF	FH	JF
1999	0.9	2.2	11.5	1.3	0.5	0.1	24.6	28.3	712	684	77.5	73.2
2000	1.0	1.2	15.0	3.1	0.6	0.1	26.1	24.0	712	643	78.2	78.0
2001	1.0	1.3	21.7	1.2	0.7	0.1	32.0	22.9	716	693	75.5	71.2
2002	1.1	1.7	27.4	1.5	0.8	0.1	36.7	25.4	718	711	73.8	68.3
2003	1.3	0.9	29.9	7.2	1.1	0.3	38.1	25.2	725	723	72.2	67.0
2004	2.4	1.9	34.9	18.3	2.1	1.1	40.5	33.1	718	720	73.6	70.6
2005	4.9	5.2	43.0	32.2	4.6	4.2	46.0	39.2	724	721	72.2	73.0
2006	6.7	7.2	46.4	38.1	6.4	6.4	48.9	43.0	723	722	72.9	75.0
2007	7.3	7.3	44.1	38.3	6.9	6.5	46.7	43.3	723	722	74.2	75.8
2008	3.5	7.5	39.4	31.5	3.3	5.5	42.4	42.6	741	730	71.9	77.9
2009	0.3	1.5	31.2		0.3	1.5	34.0		763	763	67.1	50.1
2010	0.1		25.1		0.1		28.6		762	768	69.9	63.6
2011	0.0		19.9		0.0		23.4		763	768	71.1	66.0
2012	0.0		15.8		0.0		20.2		766	772	71.3	66.6

Source: Freddie Mac, Loan Performance, Credit Suisse

Note: FH columns are Freddie Mac loans. JF columns are non-agency prime jumbo fixed loans with the following filters that make it comparable to Freddie Mac cohort: 30-year fixed rate mortgage, fully amortizing, full documentation, first-lien.

Exhibit 7: Cumulative default and severity as of December 31 2013 (60<LTV<=80, Loan Size>150k), Freddie Mac vs prime jumbo

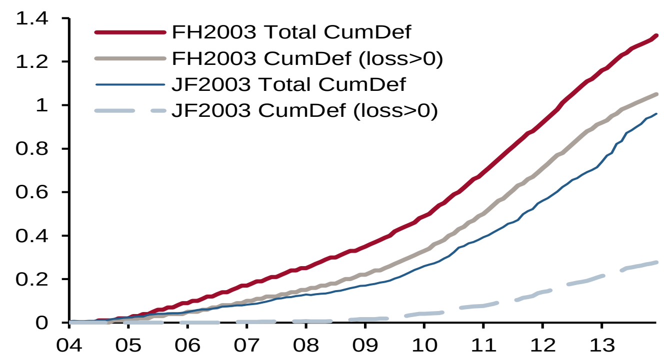
Vintage	Total CumDef		Total CumSev		CumDef (Loss>0)		CumSev (Loss>0)		FICO		LTV	
	FH	JF	FH	JF	FH	JF	FH	JF	FH	JF	FH	JF
1999	0.3	2.0	9.2	1.3	0.2	0.1	20.4	30.6	717	691	75.7	75.2
2000	0.3	1.0	13.0	3.2	0.2	0.1	22.9	24.5	719	649	76.3	78.3
2001	0.3	1.2	16.7	1.3	0.2	0.1	25.9	23.1	719	690	75.4	74.2
2002	0.4	1.7	24.3	1.7	0.3	0.1	32.5	25.8	720	709	75.1	73.6
2003	0.9	1.1	25.7	8.1	0.7	0.4	33.3	25.7	725	721	74.6	73.2
2004	2.0	2.2	33.9	18.6	1.8	1.3	38.3	31.5	720	720	75.4	74.4
2005	5.4	5.9	44.8	33.1	5.2	5.0	46.9	39.3	724	721	75.5	75.2
2006	7.4	7.9	48.7	38.8	7.2	7.1	50.6	42.8	726	723	76.0	76.2
2007	7.2	7.9	47.4	38.6	6.9	7.1	49.4	42.9	727	724	75.7	76.5
2008	3.2	6.9	42.6	30.7	3.1	4.7	45.0	45.3	742	733	74.9	75.5
2009	0.4	4.6	33.0		0.4	4.6	35.1		762	762	74.3	68.3
2010	0.1		25.7		0.1		28.3		762	767	74.9	72.7
2011	0.0		24.2		0.0		30.1		764	770	75.0	74.3
2012	0.0		18.1		0.0		27.5		767	772	74.5	73.2

Source: Freddie Mac, Loan Performance, Credit Suisse

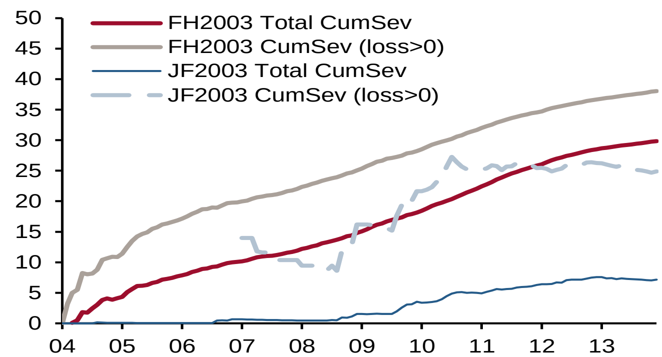
Note: FH columns are Freddie Mac loans. JF columns are non-agency prime jumbo fixed loans with the following filters that make it comparable to Freddie Mac cohort: 30-year fixed rate mortgage, fully amortizing, full documentation, first-lien.

Exhibit 8: Performance comparisons between Freddie Mac and prime jumbo fixed loans, 2003 vintage

Cumulative Default



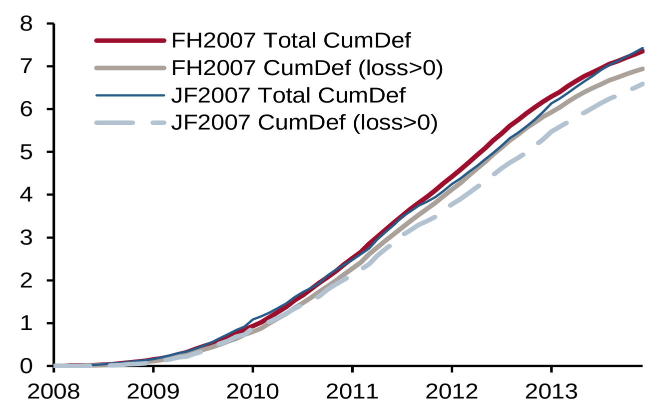
Cumulative Severity



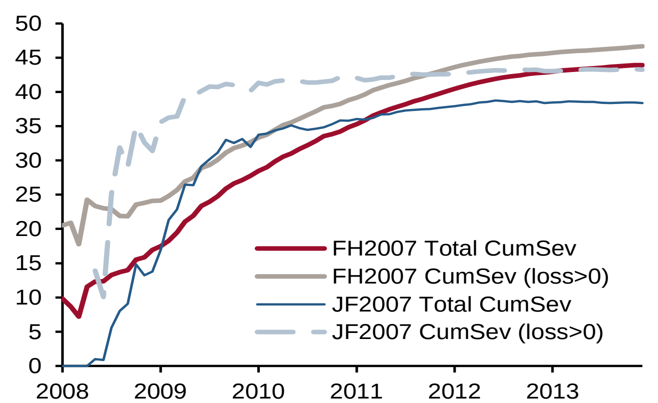
Source: Freddie Mac, Credit Suisse

Exhibit 9: Performance comparisons between Freddie Mac and prime jumbo fixed loans, 2007 vintage

Cumulative Default



Cumulative Severity



Source: Freddie Mac, Credit Suisse

Loss components

Detailed reporting gives insight to loss and severity analysis

Freddie Mac data reveals individual loss components, allowing us to analyze loss severity attributions with detail. For this analysis we calculate total loss as Default UPB + Delinquent Interest + Expenses – Net Sales Proceeds – MI Recoveries – Non MI Recoveries. Exhibits **10** and **11** show that >80 LTV loans have a lower severity than 60-80 LTV despite the lower recovery from net sales, mainly due to MI recoveries.

Exhibit 10: Loss components, % of defaulted UPB (60<LTV<=80)

Vintage	Lost Interest	Expenses	Net Sales	MI recoveries	Non-MI recoveries	Total CumSev
1999	10.3	11.0	-87.9	-0.9	-14.1	18.4
2000	11.6	11.0	-78.2	-0.2	-18.7	25.5
2001	10.1	11.4	-74.4	-0.2	-13.8	33.0
2002	9.7	11.7	-70.9	-0.3	-10.9	39.4
2003	8.6	11.3	-79.6	0.0	-5.1	35.2
2004	8.7	9.8	-73.7	0.0	-5.0	39.8
2005	8.3	7.3	-64.3	0.0	-4.1	47.2
2006	9.2	6.7	-59.7	0.0	-5.3	50.8
2007	9.2	6.8	-59.3	0.0	-6.4	50.2
2008	9.1	6.7	-62.1	0.0	-7.4	46.3
2009	6.4	6.6	-72.8	0.0	-4.4	35.7
2010	5.7	7.1	-79.2	0.0	-4.0	29.6
2011	4.8	6.3	-78.1	0.0	-3.0	29.9
2012	3.1	4.0	-91.6	0.0	-0.8	14.7

Source: Freddie Mac, Credit Suisse

Exhibit 11: Loss components, % of defaulted UPB (80<LTV<=97)

Vintage	Lost Interest	Expenses	Net Sales	MI recoveries	Non-MI recoveries	Total CumSev
1999	10.0	10.2	-80.1	-22.5	-10.6	7.0
2000	11.4	10.7	-71.7	-24.4	-15.9	10.0
2001	10.0	10.8	-66.9	-22.6	-16.6	14.7
2002	9.6	11.3	-64.0	-22.3	-16.3	18.3
2003	8.7	11.9	-66.5	-20.3	-9.6	24.3
2004	9.1	11.5	-63.9	-20.7	-8.3	27.6
2005	9.2	9.8	-57.1	-21.4	-7.8	32.6
2006	9.9	8.8	-53.3	-20.8	-9.5	35.2
2007	9.9	8.0	-52.6	-18.2	-11.1	36.0
2008	9.2	7.4	-57.0	-16.9	-12.1	30.6
2009	5.7	6.4	-71.3	-16.1	-6.2	18.5
2010	4.8	6.4	-76.8	-18.4	-3.5	12.4
2011	4.2	5.7	-78.5	-19.6	-1.3	10.5
2012	3.6	4.3	-75.4	-14.7	-0.4	17.3

Source: Freddie Mac, Credit Suisse

Credit performance by FICO and LTV buckets

>80 LTV bucket has the lowest severity of the three LTV buckets across all FICO ranges due to MI recoveries

Exhibits 12 and 13 show comparisons of cumulative severity and default as of December 2013 by FICO and LTV buckets. FICO is a significant driver for default, but not for severity. Average MI % depends strongly on LTV but weakly on FICO; >80 LTV loans' MI coverage is consistent at about 25% across vintages and FICO ranges (Exhibit 14).

Exhibit 12: Total CumSev by vintage, FICO and LTV

Vintage	0<FICO<=700			700<FICO<=760			760<FICO		
	<=60 LTV	(60,80] LTV	(80,97] LTV	<=60 LTV	(60,80] LTV	(80,97] LTV	<=60 LTV	(60,80] LTV	(80,97] LTV
1999	12.3	18.4	6.8	18.2	18.0	7.4	23.7	20.8	12.5
2000	24.3	25.4	10.2	20.8	23.5	9.6	31.3	33.8	10.6
2001	24.9	33.1	14.8	26.7	33.2	14.1	21.0	31.4	17.4
2002	29.5	40.4	18.0	21.4	37.3	19.9	29.1	38.4	21.9
2003	21.3	36.3	23.0	19.6	33.8	26.5	20.4	33.6	29.3
2004	25.8	40.7	26.3	20.9	38.7	30.4	24.6	38.4	31.0
2005	36.3	48.2	31.7	33.6	46.6	33.4	32.4	45.4	35.4
2006	43.4	52.4	34.7	39.9	50.0	36.3	37.6	47.9	35.2
2007	46.5	52.4	36.1	40.3	49.0	36.5	37.7	46.5	34.9
2008	38.9	49.5	32.3	33.0	45.0	29.6	29.3	41.9	28.1
2009	25.4	40.4	20.1	25.2	35.3	17.9	21.4	32.8	18.3
2010	29.4	32.6	16.5	29.2	28.4	12.2	17.6	29.0	10.6
2011		30.0	8.9		29.2	9.5		30.4	12.9
2012		11.2			15.7	20.1		14.1	5.7

Source: Freddie Mac, Credit Suisse

Exhibit 13: Total CumDef by vintage, FICO and LTV

Vintage	0<FICO<=700			700<FICO<=760			760<FICO		
	<=60 LTV	(60,80] LTV	(80,97] LTV	<=60 LTV	(60,80] LTV	(80,97] LTV	<=60 LTV	(60,80] LTV	(80,97] LTV
1999	0.4	1.2	2.7	0.1	0.3	0.8	0.0	0.1	0.4
2000	0.3	1.2	3.0	0.1	0.2	0.9	0.0	0.1	0.5
2001	0.3	1.2	3.9	0.1	0.3	1.1	0.0	0.1	0.6
2002	0.4	1.5	4.4	0.1	0.5	1.6	0.0	0.2	0.8
2003	0.6	2.1	4.7	0.2	0.9	2.4	0.1	0.4	1.3
2004	1.1	3.5	6.6	0.4	1.8	3.6	0.2	0.9	2.2
2005	2.4	7.8	11.4	1.2	4.9	7.8	0.4	2.4	5.0
2006	3.4	10.4	14.4	1.7	7.1	10.0	0.6	3.5	6.2
2007	3.0	10.1	15.8	1.6	7.1	11.5	0.5	3.5	7.3
2008	1.8	6.4	11.7	0.7	3.6	6.9	0.2	1.5	4.0
2009	0.3	1.3	2.7	0.1	0.5	1.2	0.0	0.2	0.6
2010	0.1	0.3	0.7	0.0	0.2	0.3	0.0	0.1	0.2
2011	0.0	0.1	0.3	0.0	0.0	0.1	0.0	0.0	0.1
2012	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Source: Freddie Mac, Credit Suisse

Exhibit 14: Average MI % by FICO, LTV buckets weighted by OrigBal

Vintage	0<FICO<=700			700<FICO<=760			760<FICO		
	<=60 LTV	(60,80] LTV	(80,97] LTV	<=60 LTV	(60,80] LTV	(80,97] LTV	<=60 LTV	(60,80] LTV	(80,97] LTV
1999	0.0	1.8	25.5	0.0	1.5	24.8	0.0	1.3	24.2
2000	0.0	0.1	25.6	0.0	0.1	24.5	0.0	0.1	24.0
2001	0.0	0.1	25.2	0.0	0.1	24.5	0.0	0.0	24.0
2002	0.0	0.1	25.4	0.0	0.1	24.9	0.0	0.0	24.3
2003	0.0	0.0	24.6	0.0	0.0	24.0	0.0	0.0	23.4
2004	0.0	0.0	24.3	0.0	0.0	24.2	0.0	0.0	23.8
2005	0.0	0.0	24.3	0.0	0.0	24.3	0.0	0.0	24.2
2006	0.0	0.0	24.3	0.0	0.0	24.3	0.0	0.0	24.3
2007	0.0	0.0	24.3	0.0	0.0	24.5	0.0	0.0	24.6
2008	0.0	0.0	24.1	0.0	0.0	23.9	0.0	0.0	23.3
2009	0.0	0.0	23.9	0.0	0.0	22.5	0.0	0.0	21.7
2010	0.0	0.0	25.8	0.0	0.0	25.0	0.0	0.0	24.5
2011	0.0	0.0	26.3	0.0	0.0	25.9	0.0	0.0	25.5
2012	0.0	0.0	26.1	0.0	0.0	26.1	0.0	0.0	25.5
2013	0.0	0.0	26.3	0.0	0.0	26.1	0.0	0.0	25.6

Source: Freddie Mac, Credit Suisse

Loss severity driver: judicial/non-judicial

Impact: delinquent interest and expenses

Judicial states' severity is higher because judicial states have a longer liquidation timeline and thus incur larger lost interest and expenses.

Exhibits 15 and 16 show cumulative loss components as a percentage of defaulted unpaid principal balance as well as default and severity across vintages.

Exhibit 15: Loss components, percentage of defaulted UPB by judicial/non-judicial, 60<LTV<=80

Vintage	Lost Interest		Expenses		Net Sales		MI recoveries		Non-MI		Total CumSev		Total CumDef	
	Judicial	Non-Jud	Judicial	Non-Jud	Judicial	Non-Jud	Judicial	Non-Jud	Judicial	Non-Jud	Judicial	Non-Jud	Judicial	Non-Jud
1999	12.5	9.2	14.5	9.1	-85.3	-89.3	-0.9	-0.9	-14.6	-13.9	26.2	14.2	0.6	0.6
2000	13.7	10.4	14.1	9.2	-72.7	-81.5	-0.2	-0.1	-20.0	-18.0	35.0	20.0	0.6	0.5
2001	11.9	9.1	14.3	9.8	-68.5	-77.7	-0.2	-0.2	-15.4	-12.9	42.0	28.1	0.6	0.6
2002	11.7	8.6	15.0	9.9	-67.9	-72.7	-0.4	-0.2	-11.8	-10.3	46.6	35.3	0.8	0.7
2003	10.7	7.7	15.3	9.4	-79.1	-79.8	0.0	0.0	-5.4	-4.9	41.5	32.4	0.9	1.2
2004	10.9	7.7	13.6	8.1	-73.6	-73.8	0.0	0.0	-5.8	-4.6	45.1	37.4	1.7	2.4
2005	10.6	7.4	10.5	6.0	-64.0	-64.5	0.0	0.0	-4.8	-3.9	52.3	45.1	3.9	5.9
2006	11.5	8.1	9.2	5.6	-58.0	-60.4	0.0	0.0	-6.0	-5.0	56.7	48.2	5.5	8.1
2007	11.7	8.2	9.5	5.7	-56.5	-60.5	0.0	0.0	-7.1	-6.2	57.6	47.3	5.1	8.0
2008	11.6	8.1	9.4	5.7	-57.8	-63.8	0.0	0.0	-7.4	-7.4	55.9	42.6	2.6	3.7
2009	7.6	5.8	8.3	5.8	-69.0	-74.6	0.0	0.0	-4.7	-4.3	42.3	32.8	0.4	0.4
2010	6.4	5.3	8.6	6.4	-76.7	-80.3	0.0	0.0	-3.7	-4.2	34.6	27.2	0.1	0.1
2011	4.4	4.9	5.4	6.6	-86.5	-75.5	0.0	0.0	-0.9	-3.7	22.4	32.3	0.0	0.0
2012	2.7	3.2	1.3	4.7	-75.5	-96.1	0.0	0.0	-0.3	-0.9	28.1	10.8	0.0	0.0

Source: Freddie Mac, Credit Suisse

Exhibit 16: Loss components, percentage of defaulted UPB by judicial/non-judicial, 80<LTV<=97

Vintage	Lost Interest		Expenses		Net Sales		MI recoveries		Non-MI		Total CumSev		Total CumDef	
	Judicial	Non-Jud	Judicial	Non-Jud	Judicial	Non-Jud	Judicial	Non-Jud	Judicial	Non-Jud	Judicial	Non-Jud	Judicial	Non-Jud
1999	11.8	9.0	13.2	8.7	-80.0	-80.1	-22.3	-22.6	-12.7	-9.6	10.1	5.4	1.6	1.8
2000	13.4	10.4	12.9	9.4	-71.9	-71.7	-22.9	-25.2	-17.9	-14.9	13.7	8.1	1.7	2.1
2001	11.6	9.1	13.0	9.5	-64.6	-68.3	-22.8	-22.5	-18.6	-15.5	18.5	12.3	2.3	2.6
2002	11.1	8.6	13.6	9.8	-63.0	-64.6	-23.0	-21.8	-17.2	-15.7	21.5	16.3	2.7	3.3
2003	10.3	7.7	14.8	10.1	-66.8	-66.3	-21.1	-19.8	-9.8	-9.5	27.5	22.2	2.9	3.7
2004	10.7	7.9	14.0	9.6	-63.5	-64.2	-21.5	-20.2	-8.5	-8.1	31.1	24.9	4.4	5.6
2005	10.8	7.7	11.6	8.1	-55.9	-58.2	-22.1	-20.8	-8.3	-7.5	36.2	29.4	8.5	9.7
2006	11.5	8.6	10.3	7.6	-51.5	-54.8	-21.1	-20.6	-10.2	-8.9	39.0	31.9	10.6	12.2
2007	12.1	8.8	10.2	6.9	-50.5	-53.8	-18.5	-18.1	-11.8	-10.7	41.5	33.1	10.0	14.7
2008	11.5	8.2	9.4	6.5	-52.0	-59.2	-16.8	-16.9	-13.4	-11.6	38.8	26.9	5.7	8.5
2009	6.4	5.4	7.2	6.0	-71.1	-71.4	-17.1	-15.7	-5.2	-6.7	20.2	17.6	0.8	1.2
2010	5.1	4.6	7.7	5.6	-73.9	-78.5	-20.8	-17.0	-2.6	-4.1	15.4	10.6	0.2	0.3
2011	4.4	4.1	5.6	5.7	-78.4	-78.5	-17.5	-20.4	-0.6	-1.6	13.4	9.4	0.1	0.1
2012	3.6	3.6	3.4	5.1	-76.3	-74.5	-18.9	-10.7	-0.2	-0.6	11.5	22.9	0.0	0.0

Source: Freddie Mac, Credit Suisse

Freddie Mac loans went through delinquency to REO much faster than their prime jumbo counterpart.

Exhibits 17 and 18 show the liquidation timeline for Freddie Mac loans and prime jumbo loans, across performance years. Although jumbo prime loans go through short sale faster than Freddie Mac loans, they take considerably longer to go through delinquency to REO. The timeline from REO to liquidation for jumbo prime loans and Freddie Mac loans are similar.

Exhibit 17: Freddie Mac average month in delinquency by performance year, delinquency state transition, and foreclosure type, weighted by balance

Factor Year	Delinq to Liquidation via Short Sale, etc.			Delinquency to REO			REO to Liquidation		
	Judicial	Non-Jud	All	Judicial	Non-Jud	All	Judicial	Non-Jud	All
2000	7.8	6.2	7.4	8.6	6.8	7.9	3.1	4.3	3.7
2001	8.4	7.0	7.8	10.0	7.8	8.7	4.0	4.9	4.9
2002	8.7	7.1	8.2	10.4	7.8	8.9	4.1	4.5	4.5
2003	9.2	7.3	8.5	11.2	7.5	8.6	4.0	4.9	4.7
2004	8.8	7.0	8.4	11.1	7.3	8.9	4.2	5.1	4.9
2005	10.1	7.6	8.9	11.2	7.4	8.8	5.0	5.7	5.5
2006	9.8	7.7	9.1	11.4	7.5	9.1	5.0	5.8	5.8
2007	9.8	7.3	8.4	11.5	7.1	8.8	5.0	6.2	5.9
2008	9.5	7.0	8.2	11.8	7.3	8.8	5.1	6.1	6.0
2009	10.7	8.0	9.2	13.4	8.5	10.1	5.1	5.7	5.7
2010	12.1	9.8	10.7	15.0	9.6	11.6	5.2	5.3	5.5
2011	13.2	10.7	11.9	16.2	10.7	12.7	6.9	6.3	6.8
2012	15.8	10.8	13.3	18.5	11.6	14.3	7.1	6.7	7.3
2013	18.0	12.5	16.2	20.1	12.0	16.6	7.3	7.2	8.0

Source: Freddie Mac, Credit Suisse

Exhibit 18: Jumbo prime average month in delinquency by factor year, delinquency state transition, and foreclosure type, weighted by balance

Year	Delinq to Liquidation via Short Sale, etc.			Delinquency to REO			REO to Liquidation		
	Judicial	Non-Jud	All	Judicial	Non-Jud	All	Judicial	Non-Jud	All
2000	2.8	2.6	2.7	10.0	7.5	7.7		3.2	3.2
2001	2.1	2.4	2.3	10.8	8.1	8.5	5.2	3.6	4.0
2002	3.3	2.8	3.0	13.8	9.2	9.9	4.4	4.8	4.8
2003	6.4	3.6	4.6	12.6	9.9	10.3	5.8	5.9	5.9
2004	5.7	5.2	5.3	16.4	10.3	11.4	4.8	6.8	6.4
2005	4.7	4.9	4.8	18.0	11.6	12.9	6.2	8.5	8.2
2006	3.3	5.4	4.5	20.5	11.8	14.5	7.3	6.7	6.8
2007	6.5	2.7	4.1	16.4	8.5	10.7	3.8	6.2	5.3
2008	4.9	3.7	4.1	15.4	8.8	9.7	5.7	4.8	4.9
2009	7.1	5.3	5.7	15.5	9.7	10.6	5.5	5.0	5.1
2010	9.8	8.0	8.4	19.6	12.9	14.1	5.2	4.3	4.4
2011	10.7	11.1	11.0	23.0	16.4	17.6	5.6	4.9	5.0
2012	13.3	13.0	13.0	29.0	18.9	21.0	5.8	6.0	6.0
2013	17.5	12.7	13.6	36.3	23.8	28.5	5.9	6.2	6.1

Source: Freddie Mac, Credit Suisse

Loss severity driver: LTV

Impact: MI recoveries and net sales proceeds

Severities for original LTV >80 loans are reduced by 20% on average because of MI recoveries. Higher updated LTV means lower recoveries from net sales proceeds and higher severity.

Exhibits 19 and 20 show cumulative loss components as percentage of defaulted unpaid principal balance as well as default and severity across vintages.

Exhibit 19: Loss components, percentage of defaulted UPB (Total Loss) by LTV bucket

Vintage	Lost Interest			Expenses			Net Sales			MI recoveries			Non-MI			Total CumSev			Total CumDef		
	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]
1999	8.9	10.3	10.0	10.3	11.0	10.2	-101.2	-87.9	-80.1	-0.1	-0.9	-22.5	-3.1	-14.1	-10.6	14.8	18.4	7.0	0.1	0.6	1.7
2000	10.2	11.6	11.4	11.8	11.0	10.7	-95.1	-78.2	-71.7	0.0	-0.2	-24.4	-3.5	-18.7	-15.9	23.4	25.5	10.0	0.1	0.5	2.0
2001	9.8	10.1	10.0	13.1	11.4	10.8	-92.3	-74.4	-66.9	-0.5	-0.2	-22.6	-5.1	-13.8	-16.6	25.0	33.0	14.7	0.1	0.6	2.5
2002	9.2	9.7	9.6	13.0	11.7	11.3	-90.0	-70.9	-64.0	-0.4	-0.3	-22.3	-4.6	-10.9	-16.3	27.2	39.4	18.3	0.1	0.7	3.0
2003	8.6	8.6	8.7	12.2	11.3	11.9	-96.6	-79.6	-66.5	0.0	0.0	-20.3	-3.7	-5.1	-9.6	20.6	35.2	24.3	0.2	1.1	3.4
2004	9.3	8.7	9.1	11.0	9.8	11.5	-93.0	-73.7	-63.9	0.0	0.0	-20.7	-3.1	-5.0	-8.3	24.3	39.8	27.6	0.5	2.2	5.0
2005	8.8	8.3	9.2	8.5	7.3	9.8	-80.3	-64.3	-57.1	0.0	0.0	-21.4	-2.1	-4.1	-7.8	34.8	47.2	32.6	1.2	5.1	9.1
2006	10.1	9.2	9.9	8.6	6.7	8.8	-73.6	-59.7	-53.3	0.0	0.0	-20.8	-3.6	-5.3	-9.5	41.5	50.8	35.2	1.7	7.0	11.4
2007	10.4	9.2	9.9	9.3	6.8	8.0	-71.7	-59.3	-52.6	0.0	0.0	-18.2	-4.8	-6.4	-11.1	43.1	50.2	36.0	1.5	6.9	12.6
2008	10.3	9.1	9.2	8.6	6.7	7.4	-77.9	-62.1	-57.0	0.0	0.0	-16.9	-5.6	-7.4	-12.1	35.4	46.3	30.6	0.6	3.3	7.4
2009	6.9	6.4	5.7	7.9	6.6	6.4	-84.8	-72.8	-71.3	0.0	0.0	-16.1	-5.7	-4.4	-6.2	24.3	35.7	18.5	0.1	0.4	1.0
2010	5.2	5.7	4.8	9.6	7.1	6.4	-88.6	-79.2	-76.8	0.0	0.0	-18.4	-1.5	-4.0	-3.5	24.7	29.6	12.4	0.0	0.1	0.3
2011		4.8	4.2		6.3	5.7		-78.1	-78.5		0.0	-19.6		-3.0	-1.3		29.9	10.5	0.0	0.0	0.1
2012		3.1	3.6		4.0	4.3		-91.6	-75.4		0.0	-14.7		-0.8	-0.4		14.7	17.3	0.0	0.0	0.0

Source: Freddie Mac, Credit Suisse

Exhibit 20: Loss components, percentage of defaulted UPB (Loss>0) by LTV bucket

Vintage	Lost Interest			Expenses			Net Sales			MI recoveries			Non-MI			CumSev (Loss>0)			CumDef (Loss>0)		
	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]	<=60	(60,80]	(80,97]
1999	11.9	12.2	11.7	24.2	15.5	13.7	-80.1	-77.7	-67.1	-0.3	-1.2	-30.2	-6.5	-14.6	-10.8	49.2	34.1	17.4	0.0	0.3	0.8
2000	14.4	13.1	12.9	27.8	14.0	12.6	-68.1	-68.0	-59.2	0.0	-0.2	-28.1	-6.9	-19.8	-19.2	67.3	39.1	19.0	0.0	0.4	1.2
2001	12.1	11.0	11.0	22.7	13.7	12.4	-74.8	-66.2	-57.4	-0.9	-0.2	-24.5	-7.3	-13.7	-18.3	51.8	44.5	23.1	0.1	0.4	1.7
2002	10.7	10.3	10.3	21.4	13.4	12.6	-73.5	-65.4	-57.5	-0.7	-0.3	-23.5	-5.7	-9.2	-15.8	52.1	48.9	26.2	0.1	0.6	2.2
2003	9.9	9.1	9.3	19.0	12.9	13.2	-81.5	-74.4	-62.3	0.0	0.0	-21.2	-5.0	-4.3	-7.6	42.4	43.3	31.3	0.1	0.9	2.7
2004	10.1	9.0	9.6	14.3	10.4	12.4	-81.5	-70.8	-60.9	0.0	0.0	-21.3	-3.8	-4.1	-6.5	39.1	44.6	33.3	0.3	1.9	4.2
2005	9.1	8.4	9.6	9.5	7.5	10.2	-74.5	-63.4	-55.2	0.0	0.0	-21.6	-2.3	-3.2	-6.3	41.8	49.3	36.7	1.0	4.9	8.1
2006	10.3	9.3	10.3	9.2	6.8	9.2	-69.5	-59.3	-52.0	0.0	0.0	-21.1	-3.5	-4.1	-8.0	46.6	52.7	38.4	1.5	6.8	10.5
2007	10.6	9.3	10.3	10.0	6.9	8.4	-67.4	-58.9	-51.7	0.0	0.0	-18.4	-4.5	-5.2	-9.6	48.7	52.1	39.0	1.3	6.6	11.7
2008	10.8	9.2	9.5	9.8	6.9	7.8	-72.1	-61.4	-55.8	0.0	0.0	-17.1	-5.7	-6.1	-10.8	42.8	48.6	33.6	0.5	3.1	6.7
2009	7.3	6.5	6.0	9.7	6.8	6.8	-77.6	-71.7	-69.5	0.0	0.0	-16.5	-6.5	-3.7	-5.7	32.9	37.9	21.1	0.0	0.4	0.9
2010	5.5	5.7	5.1	11.9	7.5	7.2	-80.9	-77.4	-74.6	0.0	0.0	-18.9	-1.7	-3.4	-3.0	34.8	32.5	15.7	0.0	0.1	0.2
2011		4.9	4.2		6.7	6.0		-75.6	-75.7		0.0	-20.0		-2.8	-1.4		33.2	13.1	0.0	0.0	0.1
2012		2.7	3.6		3.6	4.3		-81.8	-75.4		0.0	-14.7		-0.9	-0.4		23.6	17.3	0.0	0.0	0.0

Source: Freddie Mac, Credit Suisse

Loss severity driver: loan size

Impact: expenses and net sales proceeds

Because of fixed costs, a small loan size means it is more expensive to maintain and has less recoveries from net sales proceeds thus a higher severity.

Exhibits 21 and 22 show cumulative loss components as a percentage of defaulted unpaid principal balance as well as default and severity across vintages.

Exhibit 21: Loss components, percentage of defaulted UPB by loan size bucket, 60<LTV<=80

Vintage	Lost Interest			Expenses			Net Sales			MI recoveries			Non-MI recoveries			Total CumSev			Total CumDef		
	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K
1999	11.1	10.3	9.8	16.5	10.6	6.7	-72.9	-91.2	-94.7	-0.9	-0.9	-0.7	-19.1	-13.1	-11.9	34.8	15.7	9.2	2.2	0.7	0.3
2000	12.6	11.5	10.9	17.0	10.1	7.0	-59.3	-80.8	-91.6	-0.4	0.0	-0.1	-25.5	-18.2	-13.2	44.3	22.5	13.1	2.6	0.7	0.3
2001	10.7	10.1	9.7	16.9	11.6	7.7	-51.0	-73.2	-90.3	-0.6	-0.2	0.0	-19.6	-13.9	-10.3	56.4	34.5	16.8	3.6	0.9	0.3
2002	9.9	9.7	9.6	16.7	12.3	8.9	-46.5	-67.8	-85.6	-0.9	-0.2	0.0	-16.8	-10.7	-8.5	62.4	43.2	24.4	4.5	1.3	0.4
2003	8.9	8.6	8.6	18.7	13.3	9.0	-51.1	-72.8	-87.7	-0.1	0.0	0.0	-9.1	-5.9	-4.0	67.3	43.2	25.9	3.4	1.6	0.9
2004	9.1	9.0	8.6	19.1	13.1	7.8	-46.6	-67.5	-78.2	0.0	0.0	0.0	-10.6	-6.1	-4.1	70.9	48.4	34.0	4.8	2.5	2.0
2005	9.4	9.1	8.2	19.7	11.9	6.2	-42.9	-60.0	-65.6	0.0	0.0	0.0	-9.3	-5.1	-3.9	76.9	55.8	45.0	5.3	4.2	5.4
2006	10.1	9.9	9.0	18.8	11.3	5.7	-39.5	-55.3	-60.8	0.0	0.0	0.0	-11.0	-6.4	-5.1	78.3	59.5	48.9	6.3	5.5	7.4
2007	10.5	10.0	9.0	18.9	11.4	5.6	-32.3	-52.4	-61.2	0.0	0.0	0.0	-13.6	-8.4	-5.9	83.5	60.5	47.5	7.3	5.8	7.2
2008	9.9	9.5	9.0	18.3	11.0	5.7	-28.2	-52.2	-64.7	0.0	0.0	0.0	-11.1	-8.4	-7.2	88.9	60.0	42.8	5.9	3.5	3.2
2009	7.1	7.0	6.2	18.5	11.2	5.5	-52.2	-66.7	-74.5	0.0	0.0	0.0	-8.1	-5.7	-4.1	65.3	45.8	33.2	0.8	0.5	0.4
2010	5.4	6.2	5.5	15.4	11.2	5.2	-64.4	-75.6	-81.1	0.0	0.0	0.0	-4.4	-4.4	-3.8	51.9	37.4	25.8	0.3	0.2	0.1
2011	5.5	5.2	4.5	14.2	11.4	3.4	-58.3	-80.0	-78.3	0.0	0.0	0.0	-1.5	-3.7	-2.8	59.9	32.9	26.8	0.1	0.1	0.0
2012			2.6			2.5			-86.4			0.0			-0.6			18.1	0.0	0.0	0.0

Source: Freddie Mac, Credit Suisse

Exhibit 22: Loss components, percentage of defaulted UPB by loan size bucket, 80<LTV<=97

Vintage	Lost Interest			Expenses			Net Sales			MI recoveries			Non-MI recoveries			CumSev (Loss>0)			CumDef (Loss>0)		
	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K
1999	10.8	10.1	9.4	16.2	10.5	6.8	-69.4	-80.9	-83.4	-22.2	-22.7	-22.1	-15.5	-10.4	-8.7	19.9	6.5	2.0	4.2	1.9	1.1
2000	12.3	11.6	10.7	15.7	10.8	7.9	-59.6	-72.6	-76.3	-23.2	-24.4	-24.9	-22.8	-15.4	-13.5	22.4	10.0	3.9	5.2	2.3	1.3
2001	10.8	10.2	9.3	16.0	11.3	8.1	-49.8	-65.3	-75.7	-22.2	-22.6	-22.6	-22.6	-17.5	-13.1	32.2	16.1	6.0	6.9	3.3	1.6
2002	10.0	9.7	9.1	16.6	11.7	8.9	-47.0	-61.1	-74.3	-23.2	-22.6	-21.4	-20.4	-17.4	-13.2	36.0	20.4	9.2	7.5	4.2	1.8
2003	8.9	8.7	8.7	18.4	13.0	9.9	-46.4	-61.9	-74.4	-21.8	-20.9	-19.4	-15.0	-10.8	-7.5	44.1	28.1	17.2	6.1	4.1	2.7
2004	9.1	9.0	9.1	18.9	13.1	9.5	-43.0	-59.3	-69.5	-22.1	-21.7	-19.9	-14.7	-9.7	-6.5	48.2	31.4	22.7	6.8	5.4	4.6
2005	9.2	9.2	9.2	19.1	12.5	8.4	-39.7	-52.8	-59.3	-22.7	-22.0	-21.2	-13.6	-9.1	-7.1	52.3	37.9	29.9	7.5	7.6	9.9
2006	10.0	10.1	9.9	19.3	12.0	7.6	-35.8	-49.0	-55.0	-22.6	-21.8	-20.5	-16.7	-10.8	-8.9	54.3	40.5	33.1	8.4	9.1	12.5
2007	10.4	10.2	9.9	18.9	12.0	6.9	-32.8	-46.7	-54.4	-19.1	-19.4	-17.9	-17.7	-11.8	-10.8	59.7	44.3	33.7	8.6	9.7	13.7
2008	10.2	9.5	9.1	18.9	11.5	6.5	-31.7	-49.8	-58.5	-17.1	-18.0	-16.7	-15.8	-11.8	-12.2	64.5	41.5	28.3	5.9	6.0	7.7
2009	7.0	5.9	5.6	17.3	10.1	5.7	-49.8	-65.5	-72.5	-17.2	-17.5	-15.9	-7.9	-6.6	-6.1	49.4	26.5	16.9	1.0	0.9	1.0
2010	6.1	4.9	4.7	16.3	9.4	5.6	-72.3	-69.6	-78.4	-13.2	-23.0	-17.5	-6.8	-2.5	-3.7	30.2	19.2	10.7	0.4	0.3	0.3
2011	5.2	4.8	4.1	12.5	11.5	4.6	-79.3	-78.4	-78.5	-16.7	-23.9	-18.9	-0.6	-4.0	-0.9	21.1	10.0	10.4	0.2	0.1	0.1
2012			3.3			2.5			-83.6			-11.7			0.0			10.5	0.0	0.0	0.0

Source: Freddie Mac, Credit Suisse

Small loans generally have higher default, except for the 2005-2007 vintages. The housing downturn had pushed larger loans underwater faster.

Exhibit 23: Average current LTV by vintage and factor year, weighted by balance. The figure shows large loans in the 2005-2007 vintages are prone to default after 2009

Factor Year	2003			2004			2005			2006			2007			2008		
	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K	<=70k	(70k,150K]	>150K
1999																		
2000																		
2001																		
2002																		
2003	63.0	69.9	70.2															
2004	59.0	65.1	64.6	64.1	70.2	70.7												
2005	51.9	57.0	55.6	59.0	64.5	64.7	60.3	67.9	69.6									
2006	47.7	52.2	50.8	54.4	59.0	58.9	57.6	65.0	67.2	59.0	68.1	72.0						
2007	47.6	51.9	51.0	54.3	58.8	59.3	57.6	64.7	68.0	59.9	68.8	73.6	60.8	69.8	74.5			
2008	51.4	56.2	56.0	58.6	63.7	65.6	62.3	70.5	76.3	64.9	74.7	82.3	66.0	75.4	82.1	63.7	70.6	74.5
2009	56.1	61.6	62.1	63.8	70.0	73.6	68.0	78.0	87.2	71.2	83.1	95.6	72.1	83.2	94.2	67.3	75.4	81.1
2010	55.8	61.5	62.0	63.6	70.3	74.2	68.1	78.9	88.9	71.7	84.9	98.9	72.6	84.7	96.9	67.5	76.4	82.5
2011	57.0	63.4	64.3	65.0	72.6	77.7	69.9	82.0	93.7	73.9	88.6	104.5	74.7	88.1	102.0	69.6	79.6	87.0
2012	55.0	61.3	62.8	62.9	70.4	75.9	68.0	79.7	91.4	72.2	86.3	101.6	73.0	86.0	99.7	68.0	77.9	85.6
2013	50.4	55.7	56.4	57.9	64.1	68.2	62.9	72.7	81.9	66.9	79.1	91.3	67.9	79.1	90.2	63.1	71.3	77.7

Source: Freddie Mac, Credit Suisse

Loss severity driver: loan purpose

Impact: net sales proceeds

Refinance loans have higher severity due to lower recoveries from net sales proceeding, which is likely a result of inflated appraisals. (Refinance loans from 2006-2007 vintages also tend to have much higher default rates. see [May 23, 2013 Modeling and Analytics: Fannie Mae Credit data release.](#))

Exhibits 24 and 25 show cumulative loss components as percentage of defaulted unpaid principal balance as well as default and severity across vintages.

Exhibit 24: Loss components, percentage of defaulted UPB by purchase/refinance, 60<LTV<=80, LS>150K

Vintage	Lost Interest		Expenses		Net Sales		MI recoveries		Non-MI		Total CumSev		Total CumDef	
	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance
1999	9.4	10.0	6.7	6.7	-99.3	-93.0	-1.0	-0.6	-10.1	-12.5	5.7	10.5	0.2	0.5
2000	9.9	11.7	7.2	6.9	-98.9	-85.4	0.0	-0.1	-7.2	-18.3	11.0	14.8	0.2	0.5
2001	9.7	9.7	8.3	7.5	-99.9	-87.3	0.0	-0.1	-6.1	-11.6	11.9	18.3	0.2	0.3
2002	9.8	9.6	9.1	8.8	-92.8	-83.6	-0.1	0.0	-7.7	-8.8	18.3	26.0	0.3	0.5
2003	8.4	8.7	8.6	9.1	-89.8	-87.0	0.0	0.0	-3.2	-4.2	23.9	26.5	0.7	0.9
2004	8.3	8.7	7.6	7.9	-80.1	-77.2	0.0	0.0	-3.6	-4.4	32.2	35.0	1.5	2.3
2005	7.8	8.4	5.9	6.3	-67.0	-64.8	0.0	0.0	-3.9	-3.8	42.9	46.1	4.4	6.2
2006	8.4	9.4	5.3	5.9	-64.7	-58.4	0.0	0.0	-4.4	-5.5	44.7	51.4	5.5	9.4
2007	8.3	9.4	5.2	5.8	-67.5	-58.5	0.0	0.0	-5.7	-6.0	40.3	50.6	4.7	9.2
2008	9.4	8.9	5.9	5.6	-66.5	-64.2	0.0	0.0	-10.3	-6.3	38.6	43.9	1.7	4.2
2009	7.7	6.0	6.6	5.3	-67.5	-75.4	0.0	0.0	-9.3	-3.4	37.5	32.6	0.2	0.5
2010	5.5	5.5	6.4	5.0	-83.9	-80.6	0.0	0.0	-6.6	-3.4	21.4	26.5	0.0	0.1
2011	5.2	4.4	2.5	3.7	-74.0	-79.5	0.0	0.0	-0.5	-3.4	33.2	25.2	0.0	0.0
2012		2.2		2.0		-76.1		0.0		-0.6		27.5	0.0	0.0

Source: Freddie Mac, Credit Suisse

Exhibit 25: Loss components, percentage of defaulted UPB by purchase/refinance, 80<LTV<=97, LS>150K

Vintage	Lost Interest		Expenses		Net Sales		MI recoveries		Non-MI		Total CumSev		Total CumDef	
	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance
1999	9.4	9.4	6.9	6.6	-82.5	-85.6	-23.4	-18.9	-8.5	-9.1	1.9	2.4	1.0	1.9
2000	10.6	11.1	8.2	6.9	-76.7	-75.1	-25.9	-21.7	-12.8	-15.4	3.4	5.8	1.1	2.6
2001	9.2	9.5	8.3	7.9	-75.4	-76.0	-24.0	-21.3	-13.0	-13.2	5.1	6.8	1.2	2.0
2002	9.1	9.1	9.4	8.4	-74.3	-74.3	-21.7	-21.1	-14.5	-11.9	8.0	10.3	1.4	2.4
2003	8.9	8.5	10.0	9.7	-74.5	-74.3	-19.5	-19.4	-8.2	-6.7	16.7	17.8	2.5	2.9
2004	9.2	9.0	9.6	9.2	-69.3	-69.9	-20.9	-18.0	-6.6	-6.4	22.0	23.8	4.4	4.9
2005	9.2	9.1	8.3	8.5	-58.4	-61.1	-22.5	-18.5	-6.8	-7.8	29.7	30.2	10.2	9.4
2006	9.8	10.1	7.5	7.7	-55.3	-54.6	-21.9	-18.3	-8.6	-9.4	31.5	35.5	12.2	12.9
2007	9.7	10.0	6.8	7.1	-56.5	-52.3	-18.7	-17.1	-10.8	-10.8	30.5	36.9	11.6	16.9
2008	9.4	8.8	6.6	6.4	-59.8	-56.8	-16.8	-16.5	-13.8	-10.1	25.6	31.7	6.4	10.5
2009	5.9	5.5	6.4	5.3	-76.0	-70.5	-15.4	-16.1	-6.5	-5.9	14.4	18.2	0.6	1.7
2010	4.9	4.5	5.8	5.3	-82.4	-73.0	-17.8	-17.2	-3.2	-4.3	7.2	15.3	0.2	0.4
2011	4.4	3.6	4.8	4.2	-80.1	-76.3	-16.6	-21.9	-0.9	-0.9	11.6	8.8	0.1	0.2
2012		6.1		6.7		-71.0		-32.0		0.0		9.9	0.0	0.0

Source: Freddie Mac, Credit Suisse

Loss severity driver: occupancy

Impact: net sales proceeds

Investor property and second homes have lower recoveries from net sales proceeds than owner occupied homes, thus have a higher severity.

Exhibits 26 and 27 show cumulative loss components as a percentage of defaulted unpaid principal balance as well as default and severity across vintages.

Exhibit 26: Loss components, percentage of defaulted UPB by occupancy bucket, 60<LTV<=80, LS>150K

Vintage	Lost Interest			Expenses			Net Sales			MI recoveries			Non-MI recoveries			Total CumSev			Total CumDef		
	I	O	S	I	O	S	I	O	S	I	O	S	I	O	S	I	O	S	I	O	S
1999	11.6	9.8	9.0	6.5	6.7	5.6	-98.2	-94.5	-103.6	0.0	-0.8	0.0	-14.1	-11.9	-7.2	5.9	9.4	3.7	0.3	0.3	0.1
2000	14.3	10.6	13.3	9.2	6.9	7.5	-69.2	-93.1	-84.8	-1.1	0.0	0.0	-29.2	-12.4	-6.1	24.0	11.9	29.9	0.5	0.2	0.2
2001	10.1	9.6	10.8	7.3	7.7	7.9	-78.1	-90.7	-93.8	-0.9	0.0	0.0	-15.1	-10.1	-7.1	23.3	16.6	17.8	0.3	0.3	0.1
2002	9.0	9.6	11.5	7.3	9.0	7.4	-63.7	-86.5	-91.3	-0.3	0.0	0.0	-25.9	-7.8	-3.4	26.3	24.3	24.1	0.7	0.4	0.3
2003	8.7	8.6	8.5	7.7	9.0	7.6	-70.6	-88.2	-87.2	-0.1	0.0	0.0	-14.3	-3.7	-3.1	31.3	25.7	25.8	1.2	0.9	0.6
2004	8.6	8.6	8.7	7.6	7.8	6.9	-66.3	-78.7	-74.3	0.0	0.0	0.0	-8.5	-4.0	-2.8	41.3	33.7	38.6	2.3	2.0	1.6
2005	8.8	8.2	8.0	6.4	6.2	5.4	-56.6	-66.0	-61.1	0.0	0.0	0.0	-6.3	-3.8	-3.7	52.3	44.6	48.5	7.0	5.4	5.3
2006	9.8	9.0	8.9	6.0	5.7	5.1	-53.5	-61.3	-56.7	0.0	0.0	0.0	-7.9	-4.9	-6.5	54.3	48.6	50.9	10.8	7.4	7.0
2007	9.4	9.0	8.9	5.7	5.6	5.2	-53.9	-62.1	-57.5	0.0	0.0	0.0	-9.5	-5.5	-8.7	51.7	47.1	48.0	13.3	6.9	6.9
2008	9.8	8.9	8.9	5.8	5.7	4.8	-55.5	-66.4	-59.1	0.0	0.0	-0.1	-10.5	-6.5	-10.0	49.7	41.6	44.6	7.6	3.0	3.2
2009	8.0	6.2	5.8	6.0	5.5	4.0	-62.7	-75.2	-65.7	0.0	0.0	0.0	-9.7	-3.7	-8.2	41.6	32.8	35.9	0.4	0.4	0.4
2010	5.0	5.5	6.3	4.7	5.2	6.4	-85.0	-80.7	-86.8	0.0	0.0	0.0	-1.1	-4.1	-0.6	23.5	25.9	25.4	0.1	0.1	0.1
2011	4.8	4.4	5.6	3.4	3.6	1.7	-57.5	-80.8	-64.9	0.0	0.0	0.0	-0.8	-3.2	-0.3	49.9	24.0	42.1	0.0	0.0	0.0
2012		2.6			2.5			-86.4			0.0			-0.6			18.1		0.0	0.0	0.0

Source: Freddie Mac, Credit Suisse

Exhibit 27: Loss components, percentage of defaulted UPB by occupancy bucket, 80<LTV<=97, LS>150K

Vintage	Lost Interest			Expenses			Net Sales			MI recoveries			Non-MI recoveries			Total CumSev			Total CumDef		
	I	O	S	I	O	S	I	O	S	I	O	S	I	O	S	I	O	S	I	O	S
1999	12.0	9.4	9.8	2.4	6.9	7.4	-48.8	-83.8	-61.5	-14.0	-22.0	-28.8	-50.2	-8.4	-16.1	1.4	1.9	10.7	1.4	1.1	0.9
2000	9.5	10.7	10.5	5.9	8.0	6.2	-57.5	-76.9	-78.7	-19.8	-25.2	-20.6	-35.9	-12.7	-15.2	2.3	4.0	2.2	3.9	1.2	1.2
2001	8.2	9.3	10.3	5.1	8.1	7.7	-42.3	-76.2	-78.1	-19.4	-22.7	-18.8	-46.4	-12.5	-13.0	5.2	6.0	8.1	2.6	1.6	0.6
2002	7.9	9.2	8.5	5.5	8.9	7.6	-51.1	-74.8	-66.4	-17.7	-21.7	-12.7	-34.0	-12.5	-27.1	10.5	9.1	10.0	1.8	1.8	1.8
2003	9.1	8.7	8.5	9.4	9.9	8.1	-58.2	-74.6	-74.0	-15.5	-19.6	-16.2	-16.4	-7.4	-8.5	28.5	17.1	18.0	2.4	2.7	2.3
2004	9.4	9.1	8.6	9.5	9.6	8.0	-52.9	-70.0	-64.9	-20.1	-19.9	-18.7	-15.0	-6.4	-6.2	30.9	22.3	26.9	3.8	4.5	5.7
2005	10.5	9.1	9.1	8.7	8.4	7.5	-51.4	-60.0	-53.3	-21.5	-21.0	-22.6	-10.4	-7.0	-7.6	36.0	29.5	33.1	12.3	9.6	14.1
2006	11.0	9.9	9.8	7.8	7.7	6.9	-50.1	-55.8	-48.7	-19.3	-20.6	-19.8	-13.8	-8.5	-11.8	35.6	32.7	36.4	16.9	12.0	18.9
2007	9.8	9.9	10.2	6.6	6.9	6.7	-48.4	-55.0	-49.7	-15.0	-18.1	-17.3	-19.0	-10.2	-14.7	34.0	33.5	35.3	26.1	13.3	16.5
2008	10.1	9.1	9.4	6.8	6.5	6.7	-50.2	-59.3	-53.4	-13.6	-16.9	-15.4	-21.6	-11.4	-15.6	31.6	27.9	31.7	23.2	7.3	9.9
2009		5.6	5.6		5.7	4.5		-72.5	-65.7		-15.9	-16.3		-6.2	-3.3		16.8	24.8	0.0	1.0	1.6
2010		4.7			5.6			-78.4			-17.5			-3.7			10.7		0.0	0.3	0.0
2011		4.1			4.6			-78.5			-18.9			-0.9			10.4		0.0	0.1	0.0
2012		3.3			2.5			-83.6			-11.7			0.0			10.5		0.0	0.0	0.0

Source: Freddie Mac, Credit Suisse

Loss severity driver: SATO

Impact: non-MI recoveries

High SATO loans have higher Non-MI recoveries, perhaps because high SATO loan originations are prone to underwriting defects and more easily subject to recourse. On the other hand, high SATO loans associate with above average WAC, which leads to higher lost interest and expenses.

Exhibits 28 and 29 show cumulative loss components as a percentage of defaulted unpaid principal balance as well as default and severity across vintages.

Exhibit 28: Loss components, percentage of defaulted UPB by SATO bucket, 60<LTV<=80

Vintage	Lost Interest			Expenses			Net Sales			MI recoveries			Non-MI recoveries			Total CumSev			Total CumDef		
	<=0	(0,50]	>50	<=0	(0,50]	>50	<=0	(0,50]	>50	<=0	(0,50]	>50	<=0	(0,50]	>50	<=0	(0,50]	>50	<=0	(0,50]	>50
1999	9.5	10.5	12.2	9.8	11.3	13.0	-92.7	-88.3	-74.2	-0.9	-0.7	-1.5	-10.7	-14.8	-21.2	15.0	18.0	28.3	0.4	0.7	1.6
2000	10.2	11.2	12.7	9.5	10.6	12.2	-90.8	-84.0	-66.1	0.0	0.0	-0.3	-12.1	-16.9	-23.9	16.9	20.8	34.6	0.3	0.4	1.3
2001	9.4	10.1	11.2	10.9	11.6	12.0	-83.8	-75.5	-55.8	0.0	-0.1	-0.7	-9.2	-13.0	-23.5	27.3	33.1	43.2	0.4	0.6	2.2
2002	9.1	9.7	10.6	10.9	12.0	12.4	-82.0	-71.7	-53.9	0.0	-0.3	-0.5	-5.6	-10.4	-19.4	32.5	39.4	49.2	0.5	0.7	2.3
2003	8.2	8.8	9.6	10.7	11.4	12.8	-83.8	-78.8	-66.5	0.0	0.0	0.0	-3.6	-5.0	-10.7	31.4	36.4	45.1	1.0	1.1	1.7
2004	8.2	8.9	10.0	9.0	9.9	12.2	-78.3	-72.2	-62.7	0.0	0.0	0.0	-3.8	-5.1	-9.5	35.1	41.5	50.1	1.8	2.4	3.2
2005	7.9	8.6	9.9	6.7	7.7	9.9	-66.2	-63.1	-56.8	0.0	0.0	0.0	-3.6	-4.2	-8.3	44.8	49.0	54.7	4.5	5.8	7.4
2006	8.5	9.5	11.0	5.9	7.0	8.9	-61.9	-58.8	-53.2	0.0	0.0	0.0	-4.1	-5.8	-9.5	48.4	52.0	57.2	5.8	8.2	11.0
2007	8.2	9.4	11.2	5.8	7.0	8.7	-62.7	-59.2	-50.7	0.0	0.0	0.0	-4.6	-6.4	-11.4	46.7	50.7	57.8	5.3	7.7	12.9
2008	7.7	8.9	10.6	5.5	6.5	8.1	-68.1	-64.2	-54.6	0.0	0.0	0.0	-4.9	-7.1	-10.0	40.1	44.1	54.1	2.1	3.4	6.9
2009	5.2	6.3	8.3	5.4	6.7	8.3	-76.6	-74.5	-64.0	0.0	0.0	0.0	-3.2	-3.8	-7.5	30.9	34.7	45.1	0.3	0.4	1.2
2010	4.4	5.7	6.5	5.8	6.9	8.4	-79.7	-82.2	-72.4	0.0	0.0	0.0	-3.1	-2.6	-7.6	27.4	27.8	34.9	0.1	0.1	0.3
2011		4.8	5.0	3.1	7.0	5.7		-81.9	-72.8		0.0	0.0		-0.8	-0.6		29.0	37.3	0.0	0.0	0.1
2012																			0.0	0.0	0.0

Source: Freddie Mac, Credit Suisse

Exhibit 29: Loss components, percentage of defaulted UPB by SATO bucket, 80<LTV<=97

Vintage	Lost Interest			Expenses			Net Sales			MI recoveries			Non-MI recoveries			Total CumSev			Total CumDef		
	<=0	(0,50]	>50	<=0	(0,50]	>50	<=0	(0,50]	>50	<=0	(0,50]	>50	<=0	(0,50]	>50	<=0	(0,50]	>50	<=0	(0,50]	>50
1999	9.4	10.0	11.0	9.7	10.3	10.8	-83.2	-80.1	-74.7	-22.5	-22.7	-22.1	-8.2	-10.4	-15.3	5.3	7.1	9.8	1.2	1.8	3.6
2000	10.3	11.0	12.2	9.8	10.6	11.0	-79.0	-75.5	-65.8	-24.9	-25.0	-23.6	-10.5	-13.3	-20.3	5.7	7.8	13.6	1.3	1.6	3.6
2001	9.0	9.8	11.1	10.7	11.0	10.6	-75.5	-69.4	-57.2	-22.1	-23.1	-22.4	-10.5	-15.3	-23.1	11.5	13.0	19.0	1.7	2.1	6.7
2002	8.7	9.4	10.4	11.0	11.5	11.2	-72.6	-65.6	-55.8	-21.1	-22.8	-22.4	-11.3	-17.1	-18.6	14.8	15.4	24.9	2.2	2.6	5.9
2003	8.1	8.8	9.4	11.6	12.0	12.3	-71.3	-66.1	-59.3	-19.2	-20.7	-21.3	-7.2	-10.3	-12.1	22.1	23.7	28.9	3.0	3.4	4.4
2004	8.5	9.1	10.0	11.0	11.4	12.5	-68.0	-64.2	-56.3	-19.5	-21.0	-21.9	-6.5	-8.1	-11.7	25.5	27.1	32.5	4.3	5.1	6.3
2005	8.5	9.3	10.4	9.0	10.0	11.0	-60.1	-56.3	-51.8	-20.5	-21.9	-22.3	-6.9	-7.7	-10.8	30.1	33.4	36.5	7.8	9.6	11.6
2006	9.2	10.2	10.9	8.1	9.0	9.8	-56.5	-52.5	-49.3	-20.6	-21.0	-20.7	-7.5	-9.4	-13.9	32.6	36.2	36.9	9.6	12.1	14.5
2007	8.8	9.8	11.3	7.4	7.9	8.9	-57.0	-53.6	-46.9	-18.5	-18.7	-17.2	-7.2	-9.9	-16.8	33.5	35.5	39.3	9.8	12.4	18.5
2008	7.6	8.9	10.4	6.4	7.2	8.2	-63.5	-59.4	-50.7	-17.5	-17.6	-15.9	-7.2	-10.3	-16.9	25.8	28.8	35.1	4.9	6.9	12.0
2009	5.1	5.7	7.2	5.7	6.7	7.7	-73.1	-71.9	-64.4	-16.6	-16.0	-15.2	-4.5	-6.6	-10.1	16.6	17.9	25.2	0.9	0.9	2.1
2010	4.7	4.7	5.1	5.3	6.7	6.7	-81.4	-77.6	-67.0	-18.3	-18.3	-18.9	-2.4	-2.4	-9.8	7.8	13.1	16.1	0.2	0.3	0.6
2011	2.8	4.6	4.0	2.8	6.0	6.8	-74.1	-79.7	-77.7	-17.9	-19.0	-22.1	-1.1	-1.4	-1.3	12.6	10.4	9.6	0.1	0.1	0.2
2012																			0.0	0.0	0.0

Source: Freddie Mac, Credit Suisse

Loss severity driver: servicer

Impact: lost interest

Wells Fargo has below average loss severity due to its efficient servicing.

Exhibits 30 and 31 show cumulative loss components as percentage of defaulted UPB.

Exhibit 30: Loss components, percentage of defaulted UPB by servicer bucket, 60<LTV<=80

	Lost Interest				Expenses				Net Sales				MI recoveries				Non-MI				Total CumSev				Total CumDef			
Vintage	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO
1999	11	10	10	10	13	11	11	10	-89	-85	-87	-94	-1	0	-1	-2	-14	-20	-13	-12	21	15	21	12	1	1	1	0
2000	12		12	10	11		12	10	-88		-75	-84	0		0	0	-9		-19	-19	26		29	18	0		1	0
2001	11		10	9	12		12	10	-85		-70	-86	0		0	0	-7		-15	-9	32		37	25	0		1	0
2002	12		10	9	14		12	11	-84		-66	-83	0		0	0	-6		-13	-6	35		43	32	0		1	1
2003	10	10	9	8	12	12	12	10	-85	-80	-76	-84	0	0	0	0	-3	-7	-5	-4	34	35	39	30	1	1	1	1
2004	9	10	9	8	10	10	10	9	-70	-75	-72	-76	0	0	0	0	-3	-4	-5	-5	46	40	42	36	5	2	2	2
2005	9	9	8	8	7	7	8	7	-63	-64	-64	-65	0	0	0	0	-2	-2	-6	-3	52	50	46	47	7	5	5	5
2006	11	11	9	8	7	7	7	7	-58	-59	-60	-61	0	0	0	0	-3	-4	-7	-5	56	54	49	49	12	8	6	7
2007	10	11	9	8	7	7	7	7	-58	-58	-59	-62	0	0	0	0	-5	-5	-7	-7	55	55	49	46	12	7	6	7
2008	10	10	9	8	7	6	7	7	-62	-61	-63	-62	0	0	0	0	-6	-6	-8	-9	48	50	45	45	4	4	3	3
2009	6	6	7	6	6	6	7	7	-72	-76	-72	-75	0	0	0	0	-5	-3	-5	-4	35	33	37	33	0	0	0	0
2010	5	6	6	6	6	6	7	8	-86	-75	-78	-80	0	0	0	0	-2	-13	-4	-2	23	24	30	32	0	0	0	0
2011	8	7	5	5	12	11	6	6	-52	-81	-78	-78	0	0	0	0	-3	-1	-3	-4	64	36	30	29	0	0	0	0
2012			3	3			3	6			-94	-88			0	0			-1	-1			11	20	0	0	0	0

Source: Freddie Mac, Credit Suisse

Exhibit 31: Loss components, percentage of defaulted UPB by servicer bucket, 80<LTV<=97

	Lost Interest				Expenses				Net Sales				MI recoveries				Non-MI				Total CumSev				Total CumDef			
Vintage	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO	BOA	CHASE	OTHERS	WELLS FARGO
1999	10	10	10	9	11	10	10	10	-83	-80	-79	-82	-21	-22	-23	-23	-8	-12	-11	-10	9	7	8	4	2	2	2	1
2000	11		12	11	10		11	10	-76		-70	-75	-23		-24	-25	-11		-16	-14	11		12	7	1		2	2
2001	11		10	9	12		11	11	-73		-64	-73	-21		-23	-22	-10		-19	-10	19		16	14	2		3	2
2002	10		10	9	13		11	11	-75		-62	-70	-18		-23	-20	-6		-19	-9	25		18	20	2		3	2
2003	10	10	9	8	14	12	12	12	-69	-66	-65	-69	-8	-21	-21	-19	-6	-9	-12	-6	42	27	23	26	4	4	3	3
2004	11	10	9	9	14	12	12	11	-57	-65	-63	-65	-15	-21	-21	-21	-4	-6	-10	-7	49	30	26	28	11	5	5	5
2005	12	11	9	9	12	10	10	9	-55	-56	-58	-57	-16	-22	-22	-22	-5	-5	-11	-6	47	38	29	33	11	9	8	10
2006	13	11	10	9	11	9	9	8	-52	-52	-54	-53	-17	-21	-21	-22	-5	-7	-12	-8	49	40	32	35	20	11	10	13
2007	12	12	9	9	9	9	8	8	-52	-50	-53	-55	-15	-19	-19	-19	-8	-10	-12	-13	45	41	34	30	15	15	11	13
2008	10	10	9	8	9	7	7	7	-59	-54	-57	-58	-15	-17	-16	-18	-8	-13	-13	-13	37	33	30	25	8	8	7	7
2009	5	6	6	5	7	5	6	6	-75	-73	-70	-73	-13	-13	-17	-17	-5	-1	-6	-7	20	25	19	15	1	0	1	1
2010	5	4	5	5	7	4	6	7	-78	-83	-76	-77	-14	-12	-19	-21	-7	-1	-3	-4	13	12	13	10	0	0	0	0
2011	4	5	4	4	7	11	5	6	-84	-91	-82	-70	-16	-15	-15	-28	-2	-1	-1	-1	9	9	10	11	0	0	0	0
2012			4				4				-75				-15				0				17		0	0	0	0

Source: Freddie Mac, Credit Suisse

Loss severity driver: FICO

No strong impact to loss severity

High FICO borrowers have a lower default rate, but FICO and severity do not have strong correlation.

Exhibits 32 and 33 show cumulative loss components as percentage of defaulted UPB.

Exhibit 32: Loss components, percentage of defaulted UPB by FICO bucket, 60<LTV<=80

	Lost Interest			Expenses			Net Sales			MI recoveries			Non-MI recoveries			Total CumSev			Total CumDef		
Vintage	<=700	[700,760]	>760	<=700	[700,760]	>760	<=700	[700,760]	>760	<=700	[700,760]	>760	<=700	[700,760]	>760	<=700	[700,760]	>760	<=700	[700,760]	>760
1999	10.5	9.8	9.9	11.0	10.7	11.3	-86.9	-91.8	-87.3	-1.0	-0.5	-0.4	-15.2	-10.3	-12.6	18.4	18.0	20.8	1.2	0.3	0.1
2000	11.7	11.2	11.1	10.9	10.7	13.8	-77.2	-83.3	-80.8	-0.1	-0.3	-0.1	-20.0	-14.8	-10.3	25.4	23.5	33.8	1.2	0.2	0.1
2001	10.2	9.8	9.6	11.3	11.7	11.6	-73.1	-77.1	-81.8	-0.2	-0.3	-0.2	-15.1	-10.8	-7.9	33.1	33.2	31.4	1.2	0.3	0.1
2002	9.9	9.5	8.7	11.8	11.6	11.9	-68.7	-75.5	-75.1	-0.2	-0.4	-0.3	-12.5	-7.9	-6.8	40.4	37.3	38.4	1.5	0.5	0.2
2003	9.0	8.2	7.9	11.8	10.6	10.7	-78.5	-80.6	-81.9	0.0	0.0	0.0	-5.9	-4.3	-3.0	36.3	33.8	33.6	2.1	0.9	0.4
2004	9.2	8.2	7.6	10.5	8.9	8.6	-73.2	-74.5	-74.2	0.0	0.0	0.0	-5.8	-4.0	-3.6	40.7	38.7	38.4	3.5	1.8	0.9
2005	8.9	8.0	7.1	8.1	6.8	5.8	-64.0	-64.7	-64.3	0.0	0.0	0.0	-4.8	-3.6	-3.2	48.2	46.6	45.4	7.8	4.9	2.4
2006	9.8	8.9	7.9	7.5	6.2	5.2	-59.0	-60.0	-61.1	0.0	0.0	0.0	-6.0	-5.1	-4.0	52.4	50.0	47.9	10.4	7.1	3.5
2007	9.9	8.9	7.8	7.8	6.1	5.1	-57.9	-60.2	-61.5	0.0	0.0	0.0	-7.4	-5.8	-4.9	52.4	49.0	46.5	10.1	7.1	3.5
2008	10.1	8.7	7.5	7.9	6.3	5.2	-60.2	-62.6	-65.1	0.0	0.0	0.0	-8.3	-7.4	-5.7	49.5	45.0	41.9	6.4	3.6	1.5
2009	8.0	6.2	5.3	8.1	6.6	5.5	-69.5	-73.3	-74.8	0.0	0.0	0.0	-6.2	-4.2	-3.3	40.4	35.3	32.8	1.3	0.5	0.2
2010	6.7	5.6	5.0	8.6	6.7	6.5	-77.0	-79.3	-80.9	0.0	0.0	0.0	-5.7	-4.5	-1.6	32.6	28.4	29.0	0.3	0.2	0.1
2011	5.5	5.1	3.9	5.7	6.6	6.5	-80.6	-77.3	-76.8	0.0	0.0	0.0	-0.6	-5.2	-3.2	30.0	29.2	30.4	0.1	0.0	0.0
2012		3.0	2.6		2.4	5.3		-89.0	-93.0		0.0	0.0		-0.6	-0.8		15.7	14.1	0.0	0.0	0.0

Source: Freddie Mac, Credit Suisse

Exhibit 33: Loss components, percentage of defaulted UPB by FICO bucket, 80<LTV<=97

	Lost Interest			Expenses			Net Sales			MI recoveries			Non-MI recoveries			Total CumSev			Total CumDef		
Vintage	<=700	[700,760]	>760	<=700	[700,760]	>760	<=700	[700,760]	>760	<=700	[700,760]	>760	<=700	[700,760]	>760	<=700	[700,760]	>760	<=700	[700,760]	>760
1999	10.1	9.6	10.1	10.1	10.5	11.5	-79.5	-83.2	-77.1	-22.8	-21.3	-20.3	-11.0	-8.2	-11.1	6.8	7.4	12.4	2.7	0.8	0.4
2000	11.6	10.9	11.0	10.6	10.6	11.2	-71.6	-73.0	-73.3	-24.9	-22.2	-20.0	-15.5	-16.7	-18.2	10.2	9.6	10.6	3.0	0.9	0.5
2001	10.2	9.3	9.2	10.7	10.8	11.6	-66.2	-70.1	-67.9	-22.9	-21.2	-21.4	-16.9	-14.6	-14.0	14.8	14.1	17.4	3.9	1.1	0.6
2002	9.7	9.0	9.1	11.2	11.6	12.2	-63.5	-66.0	-64.6	-22.5	-21.8	-19.5	-16.8	-12.7	-15.3	18.0	19.9	21.8	4.4	1.6	0.8
2003	8.9	8.5	7.9	12.0	11.9	11.5	-66.5	-66.3	-65.1	-21.0	-19.0	-17.7	-10.2	-8.3	-7.2	23.0	26.4	29.3	4.7	2.4	1.3
2004	9.2	9.0	8.1	11.6	11.3	10.5	-64.3	-62.9	-61.1	-21.1	-19.6	-19.2	-8.7	-7.2	-7.1	26.2	30.3	31.0	6.6	3.6	2.2
2005	9.4	8.9	8.6	10.2	9.1	8.4	-58.0	-55.5	-54.5	-21.6	-21.2	-20.2	-8.1	-7.6	-6.6	31.6	33.3	35.3	11.4	7.8	5.0
2006	10.2	9.8	8.9	9.3	8.3	7.4	-54.1	-51.6	-51.5	-21.0	-20.4	-20.1	-9.5	-9.4	-9.0	34.6	36.1	35.0	14.4	10.0	6.2
2007	10.3	9.7	9.0	8.6	7.4	6.4	-52.5	-51.7	-53.3	-18.5	-17.7	-17.6	-11.5	-10.9	-9.2	35.9	36.4	34.7	15.8	11.5	7.3
2008	9.8	8.9	8.1	8.0	7.1	6.2	-55.3	-57.4	-59.0	-16.8	-16.8	-16.8	-13.2	-11.8	-9.8	32.2	29.4	27.9	11.7	6.9	4.0
2009	6.8	5.6	5.1	8.1	6.1	5.8	-69.3	-70.7	-72.0	-16.2	-16.2	-15.7	-8.3	-6.6	-4.2	19.9	17.9	18.1	2.7	1.2	0.6
2010	5.0	5.0	4.4	7.1	6.8	5.1	-71.4	-78.3	-73.5	-21.0	-16.5	-19.0	-1.7	-3.6	-4.2	16.3	12.0	10.4	0.7	0.3	0.2
2011	5.2	4.3	3.4	7.7	5.4	4.7	-74.5	-82.5	-73.4	-28.1	-15.1	-20.0	-1.4	-0.9	-1.8	8.9	9.3	12.9	0.3	0.1	0.1
2012		3.7	3.0		4.2	4.8		-74.8	-77.9		-12.7	-23.4		-0.3	-0.8		20.1	5.7	0.0	0.0	0.0

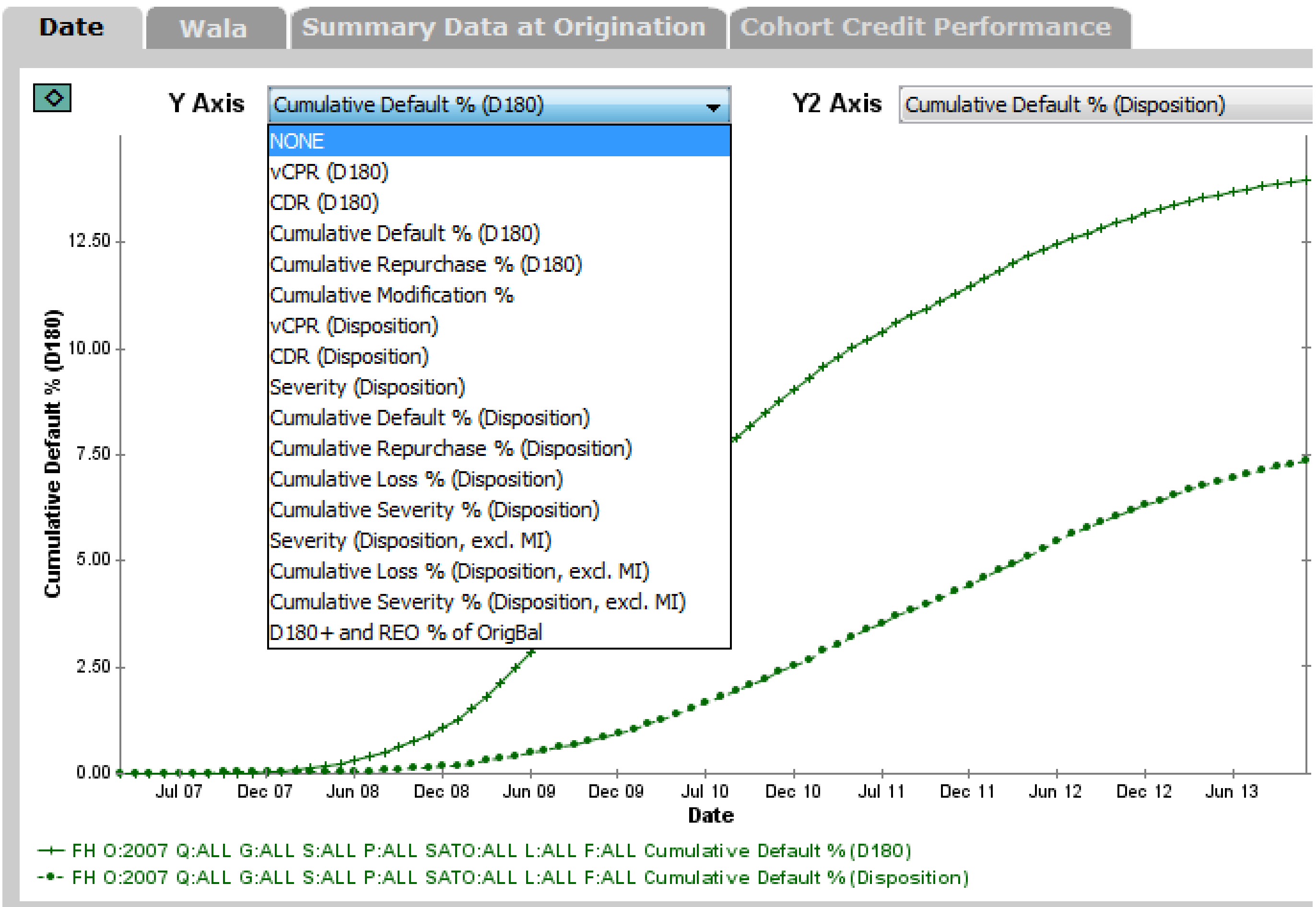
Source: Freddie Mac, Credit Suisse

Appendix: GSE Credit Analyzer on Locus updated

Existing vectors are based on D180; new vectors are based on property disposition

GSE Credit Analyzer was updated on April 6, 2015 to include Freddie Mac loss severity vectors, based on the Single Family Loan-Level Dataset released by Freddie Mac on November 24, 2014 (Exhibit **34**). This dataset includes approximately 16.9 million fully amortizing 30-year fixed-rate loans originated from January 1, 1999 through June 30, 2013, with monthly loan performance data through December 31, 2013.

Exhibit 34: New loss severity vectors in GSE Credit Analyzer



Source: Credit Suisse Locus

Freddie Mac extended loan-level credit performance data beyond 180 days' delinquency and included actual loss components through property disposition. Freddie Mac also changed loan termination event zero balance code definitions.

D180 defaults are defined as loans that become 180+ days delinquent, third-party sale prior to D180, short sale or short payoff prior to D180, Deed-in-Lieu of foreclosure prior to D180, or REO Liquidation prior to D180.

Property disposition defaults are defined as loans terminated through property disposition via either foreclosure alternative group (short sale, third-party sale, charge off and note sale) or REO disposition.

We renamed the existing vectors in the GSE Credit Analyzer with a (D180) suffix, and added the following new vectors:

- vCPR (Disposition)
- CDR (Disposition)
- Severity (Disposition)
- Cumulative Default % (Disposition)
- Cumulative Repurchase % (Disposition)
- Cumulative Loss % (Disposition)
- Cumulative Severity % (Disposition)
- Severity (Disposition, excl. MI)
- Cumulative Loss % (Disposition, excl. MI)
- Cumulative Severity % (Disposition, excl. MI)
- D180+ and REO % of OrigBal

Definitions of the new vectors

vCPR (Disposition)

Single-month loan prepayment (in full) or repurchase prior to default, corresponding to Zero Balance Code 01 or 06.

CDR (Disposition)

Single-month loan default via foreclosure alternative disposition or REO disposition, corresponding to Zero Balance Code 03 or 09.

Severity (Disposition)

Single-month loss divided by disposition default UPB. Loss is calculated as Default UPB + Delinquent Interest + Expenses – Net Sales Proceeds – MI Recoveries – Non MI Recoveries.

Cumulative Default % (Disposition)

Cumulative disposition default UPB divided by total original UPB.

Cumulative Repurchase % (Disposition)

Cumulative post disposition default event repurchase UPB divided by total original UPB.

Cumulative Loss % (Disposition)

Cumulative disposition loss amount divided by total original UPB.

Cumulative Severity % (Disposition)

Cumulative disposition loss divided by cumulative disposition default.

Severity (Disposition, excl. MI)

Same as *Severity (Disposition)*, except that MI Recoveries were excluded from the loss calculation.

Cumulative Loss % (Disposition, excl. MI)

Same as *Cumulative Loss % (Disposition)*, except that MI Recoveries were excluded from the loss calculation.

Cumulative Severity % (Disposition, excl. MI)

Same as *Cumulative Severity % (Disposition)*, except that MI Recoveries were excluded from the loss calculation.

D180+ and REO % of OrigBal

Total UPB for loans that are delinquent for 180 days or more, and loans that are in REO, divided by total original UPB.

The new vectors were added to the “Date” and “Wala” tabs, as well as the “Cohort Credit Performance” tab.

In the “Date” or “Wala” tab, Y2 Axis can be used along with Y Axis for comparison purpose. To set both axes on the same scale, right click on the chart and select “Edit Chart,” then choose “Axis Format” tab, uncheck the Max Auto box for “Y Axis” and “Y2 Axis,” enter same Max numbers, then click Apply.

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Quantitative Strategies

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